



Sprint 3 Planning Document

Team 26 BoilerBridge: Xavion Marable, Logan Wilkinson, Emilio Guzman, Shreyansh Tehanguria, Shashank Mukkera

Sprint Overview

The goal of Sprint 3 is to finish the core user experience by implementing external calendar synchronization, interactive itinerary maps, group voting mechanisms, and location review systems. This sprint will transform the application from a basic planning tool into a real-time travel companion. We are prioritizing third-party integrations (Google/Outlook APIs, mapping libraries) and making sure that data states are still handled securely and efficiently.

Scrum Master: Logan Wilkinson

Meeting Schedule: Tuesdays, Wednesdays, Thursdays @ 4:30 PM
(Discord)

Risks and Challenges

- **Third-Party API Limits** - Integrating external mapping libraries and calendar OAuth flows introduces the risk of hitting API rate limits or facing unexpected token expirations.
- **Data Synchronization Conflicts** - Ensuring that modifications to an itinerary accurately reflect on an external Google/Outlook calendar without creating duplicate events will require strict mapping schemas and robust error handling.
- **UI State Management** - Implementing visual map rendering and realtime voting tallies requires complex frontend state management to prevent visual desyncs when multiple users interact with the itinerary at the same time.

2. Current Sprint Detail

User Story #1: Group Voting

"As a user, I would like to be able to vote on specific activity options created by the itinerary generator."

- **Owner:** Xavion Marable
- **Estimated Hours:** 8.5 Hours
- **Tasks:**
 - Create a Voting schema in Mongoose to track upvotes and downvotes per activity, linked to the specific groupID and userID to prevent duplicate voting. (2 hrs)
 - Build the backend POST and DELETE routes to handle vote casting and retracting. (2 hrs)
 - Develop the Voting UI components with interactive icons to attach to individual itinerary cards. (1.5 hrs)
 - Implement real-time WebSocket or polling logic so that the group sees the winning activities as the tallies update. (1.5 hrs)
 - Write unit and integration tests to verify that users can only vote once per activity, vote totals calculate correctly, and backend routes properly authorize group members. (1.5 hrs)
- **Acceptance Criteria:**
 - **Given** a list of generated activities, **When** I click the vote button, **Then** the vote count increases globally for everyone in the group.
 - **Given** I have already voted on an activity, **When** I click the vote button again, **Then** my previous vote is retracted and the tally decreases.
 - **Given** a user is not a member of the travel group, **When** they attempt to hit the voting API endpoint, **Then** the server rejects the request with a 403 error.

User Story #2: Member Removal

"As a group leader, I would like to be able to remove specific people from the travel group."

- **Owner:** Xavion Marable
- **Estimated Hours:** 4.5 Hours
- **Tasks:**
 - Create the member removal backend logic with verification to make sure only Leaders can trigger the deletion route. (1 hr)
 - Add remove buttons to the Member List UI restricted entirely to the Leader view. (1 hr)
 - Implement a confirmation modal to prevent accidental removals from the group. (1 hr)
 - Write unit and integration tests to make sure that a removed user loses access to the group board and itinerary immediately. (1.5 hrs)
- **Acceptance Criteria:**
 - **Given** I am the group leader, **When** I click remove next to a member's name, **Then** they are immediately kicked from the group.
 - **Given** I am a regular Admin or Viewer, **When** I view the member list, **Then** the remove button is completely hidden.
 - **Given** a user is removed, **When** they try to access the group link, **Then** they are redirected to their personal dashboard.

User Story #3: Drag and Drop Reordering

"As a user, I would like to be able to drag and drop activities to reorder the schedule visually."

- **Owner:** Xavion Marable
- **Estimated Hours:** 7.5 Hours
- **Tasks:**
 - Create a drag-and-drop library for the frontend itinerary view (2.5 hrs)
 - Build logic to update the order or startTime field in the database when a card is dropped into a new position (2 hrs)
 - Implement a grid-snapping UI feel so activities align cleanly with time slots. (1.5 hrs)

- Write unit and integration tests to verify that the new order persists in the database after the drag-and-drop action. (1.5 hrs)
- **Acceptance Criteria:**
 - **Given** I want to move lunch to a later time, **When** I drag the activity card down the list, **Then** the rest of the schedule shifts to accommodate the change.
 - **Given** I am a Viewer, **When** I try to drag a card, **Then** the UI prevents the movement and shows a read-only tooltip.
 - **Given** a schedule change is made, **When** I refresh the page, **Then** the activities stay in the new order I set.

User Story #4: Lockdown Mode

"As a user, I would like to be able to lock specific activities I like so they don't change during regeneration."

- **Owner:** Xavion Marable
- **Estimated Hours:** 5.5 Hours
- **Tasks:**
 - Add an isLocked boolean field to the Activity schema in the database. (0.5 hrs)
 - Build a toggleable lock icon on the frontend activity cards. (1 hr)
 - Update the itinerary generation logic to skip over any activity where the isLocked field is true. (2.5 hrs)
 - Write unit and integration tests to verify that locked items remain in the itinerary even when a full regeneration is triggered. (1.5 hrs)
- **Acceptance Criteria:**
 - **Given** I have a dinner reservation I can't cancel, **When** I lock that activity, **Then** it remains in the schedule even if I hit a regenerated itinerary.
 - **Given** an activity is locked, **When** I try to drag it to a different day, **Then** the UI asks if I want to unlock it first.
 - **Given** I hit regenerate, **When** the new plan is created, **Then** all locked items stay in their original time slots.

User Story #5:

"As a user, I would like to be able to link my external calendar (Google/Outlook) to sync trip dates."

- **Owner:** Emilio Guzman Ochoa
- **Estimated Hours:** 18 hrs
- **Tasks:**
 - Create a `CalendarConnection` schema in Mongoose to store external calendar connection data and sync settings (`userId`, `provider`, `providerAccountId`, `calendarId`, `accessToken` or secure token reference, `refreshToken` or secure token reference, `tokenExpiresAt`, `syncEnabled`, `lastSyncedAt`, `syncError`, `createdAt`, `updatedAt`) **(2 hrs)**
 - Create a `CalendarEventSync` mapping schema to associate internal trip/calendar events with external provider event IDs so future syncs update existing events instead of creating duplicates **(1.5 hrs)**
 - Build OAuth backend flow for Google and Outlook account linking, including provider authorization, callback handling, token exchange, and secure storage of connection credentials **(3 hrs)**
 - Build POST `/api/calendar-connections` to create and save a new linked calendar connection with validation (authenticated user, supported provider, no duplicate active connection per provider for the same user) **(1.5 hrs)**
 - Build GET `/api/calendar-connections` to fetch the user's linked external calendars, selected calendar, sync status, and last sync information **(1 hr)**
 - Build PUT `/api/calendar-connections/:id` to update sync settings such as selected calendar, sync enabled/disabled, and preferences for trip date syncing **(1.5 hrs)**

- Build DELETE `/api/calendar-connections/:id` to unlink an external calendar, remove connection data, and clean up related sync mapping records **(1 hr)**
- Build provider abstraction/helper layer for Google Calendar and Outlook Calendar so sync logic can use common methods such as `listCalendars`, `createEvent`, `updateEvent`, and `deleteEvent` **(2 hrs)**
- Build backend sync logic so trip start/end dates and related itinerary events can be pushed to the linked external calendar, and existing synced events can be updated or removed when trip details change **(2.5 hrs)**
- Create External Calendar UI section/page with Google and Outlook connect buttons, linked account display, calendar selector, sync toggle, sync status, error message area, and unlink action **(2 hrs)**
- Connect UI to API and OAuth flow so users can link accounts, choose a calendar, enable sync, unlink calendars, and view loading/success/error states **(1 hr)**
- Write unit and integration tests to verify create, read, update, and delete functionality for calendar connections, authorization behavior, sync mapping behavior, duplicate prevention, and successful trip date synchronization for supported providers **(2 hrs)**
- **Acceptance Criteria:**
 - **Given** I am an authenticated user, **when** I link a supported calendar provider such as Google or Outlook, **then** the system saves the external calendar connection and shows it as linked in the UI.
 - **Given** I have linked an external calendar and enabled sync, **when** trip dates are created or updated, **then** the system synchronizes those trip dates to my selected external calendar.
 - **Given** I choose a specific calendar from my linked provider account, **when** I save my sync settings, **then** the selected calendar is stored and used for future sync operations.

- **Given** I attempt to link an unsupported provider or submit invalid connection information, **when** I complete the linking flow, **then** the system blocks the request and shows an error.
- **Given** I am not authenticated, **when** I call the calendar connection endpoints, **then** the API denies access with a 401 or 403 response.
- **Given** a calendar connection already exists, **when** I update the sync settings or selected calendar, **then** the changes are persisted and reflected in the UI.
- **Given** a calendar connection exists, **when** I unlink it, **then** the connection is removed and future trip changes are no longer synced to that external calendar.
- **Given** trip dates have already been synced once, **when** the trip dates are modified later, **then** the corresponding external calendar event is updated instead of creating a duplicate.
- **Given** a sync attempt fails because of expired credentials, revoked access, or provider API errors, **when** the system attempts to sync, **then** the user is shown an appropriate error state and the sync is not marked as successful.

User Story #6:

"As a user, I would like to be able to see a visual map representation of the itinerary."

- **Owner:** Emilio Guzman Ochoa
- **Estimated Hours:** 14.5 hrs
- **Tasks:**
 - Create a map-ready itinerary data structure or schema support so itinerary items include the location information needed for map rendering, such as place name, address, latitude, longitude, order in itinerary, tripId, and groupId (**1.5 hrs**)
 - Build GET `/api/groups/:groupId/itinerary/map` to fetch itinerary locations for a trip or group in the correct order, with validation for authenticated users and group membership (**1.5 hrs**)

- Add backend validation to ensure itinerary items without coordinates or valid location data are either excluded safely or returned with a clear fallback status for the frontend **(1 hr)**
- Integrate a mapping library such as Google Maps, Mapbox, or Leaflet into the frontend so the application can render markers and route lines for itinerary stops **(2.5 hrs)**
- Create an Itinerary Map UI section/page that displays itinerary stops as markers on the map, includes labels or popups for each stop, and visually connects the stops in itinerary order **(2.5 hrs)**
- Add map interaction features such as zooming, panning, selecting a stop from the itinerary list, and highlighting the corresponding marker on the map **(1.5 hrs)**
- Connect UI to API so the map loads trip itinerary data dynamically and updates when itinerary items are added, removed, or reordered **(1.5 hrs)**
- Add loading, empty, and error states for the map view, including cases where the itinerary has no mapped stops or location data is incomplete **(1 hr)**
- Write unit and integration tests to verify the itinerary map data endpoint, authorization rules, correct marker ordering, fallback behavior for missing coordinates, and frontend rendering of mapped itinerary stops **(2 hrs)**
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- **Acceptance Criteria:**

- **Given** I am a member of a travel group with an itinerary that includes valid location data, **when** I open the itinerary map view, **then** I can see a visual map displaying the itinerary stops as markers.
- **Given** an itinerary contains multiple stops in a planned order, **when** I view the map representation, **then** the stops are shown in the correct itinerary order and are visually connected or distinguishable in sequence.

- **Given** I select or click on a stop in the itinerary list or on the map, **when** the interaction is triggered, **then** the corresponding location is highlighted and its details are displayed clearly.
- **Given** I am not authenticated or not a member of the group, **when** I request the itinerary map data or try to open the map view, **then** the system denies access with a 401 or 403 response.
- **Given** itinerary stops are added, edited, removed, or reordered, **when** I reload or revisit the map view, **then** the map reflects the updated itinerary data accurately.
- **Given** a trip has no itinerary stops with valid location data, **when** I open the map view, **then** the UI shows an appropriate empty state instead of a broken or blank map.

User Story #7:

"As a user, I would like to be able to write reviews for places I've visited."

- **Owner:** Logan Wilkinson
- **Estimated Hours:** 10.5 hrs
- **Tasks:**
 - Review and Verify existing **POST /api/reviews** route, (1.5 hr)
 - Review and Verify existing **GET /api/reviews** route, (1.5 hr)
 - Review and Verify existing **DELETE /api/reviews** route, (1.5 hr)
 - Build and Integrate UI for reviews from locations visited. (3 hrs)
 - Build place search UI and **DELETE /api/reviews/search** route. (2.5 hrs)
 - Manual Testing and reporting to verify all features. (30 mins)
- **Acceptance Criteria:**
 - **Given**, I visit the page for a location **When**, the page loads **Then**, the reviews for the location are populated.

- **Given**, I write a review **When**, I submit it **Then**, it immediately reflects on the client and is persistent on reload.
- **Given**, when I have already written a review **When**, I click the delete button on it **Then**, it gets deleted reflected locally and on reload.

User Story #8:

"As a user, I would like to be able to share itineraries via link."

- **Owner:** Logan Wilkinson
- **Estimated Hours:** 7.5 hrs
- **Tasks:**
 - Create **GET**
`/api/group/[groupid]/groups/itinerary/link`
route. (2 hrs)
 - Create **PATCH**
`/api/group/[groupid]/groups/itinerary/link/options` route. (2 hrs)
 - Integrate toggle into itinerary UI to disable access via external link. (1 hr)
 - Create itinerary generation UI with sharing options. (2 hrs)
 - Manual Testing and reporting to verify all features. (30 mins)
- **Acceptance Criteria:**
 - **Given**, I am on my group's itinerary **When** I click the share via link button, a link is generated for me to send to friends.
 - **Given**, I am on my group's itinerary **When**, I click the sharing options button **Then**, a settings UI pops up.
 - **Given**, I am in the settings UI **When**, I change a setting **Then**, it is saved and reflects immediately.

User Story #9:

"As a user, I would like to be able to view a 10-day weather forecast in the itinerary."

- **Owner:** Logan Wilkinson

- **Estimated Hours:** 5.5 hrs
- **Tasks:**
 - Create `GET /api/groups/[groupid]/itinerary/forecast` route. (2.5 hrs)
 - Implement 10-day weather forecast UI into itinerary page. (2.5 hrs)
 - Manual Testing and reporting to verify all features. (30 mins)
- **Acceptance Criteria:**
 - **Given**, I am visiting the itinerary in a group **When**, the page loads **Then**, I will see a 10 day weather forecast for the location of the trip.
 - **Given**, I am visiting the itinerary in a group **When**, I click on the weather forecast **Then**, I will be taken to the weather website it was pulled from.
 - **Given**, there is a group itinerary already created **When**, change the location of the trip **Then**, the 10 day forecast reflects that change.

User Story #10:

"As a user, I would like to be able to receive in-app notifications about my travel."

- **Owner:** Logan Wilkinson
- **Estimated Hours:** 7.5 hrs
- **Tasks:**
 - Add settings for in app notifications in settings page. (1.5 hrs)
 - Implement travel notifications UI into existing UI. (1.5 hrs)
 - Implement `GET /api/user/me/notifications` route. (2 hrs)
 - Implement notification storage and retrieval sitewide where applicable. (2 hrs)
 - Manual Testing and reporting to verify all features. (30 mins)
- **Acceptance Criteria:**

- **Given**, when I load into the dashboard **When**, I open the notification bell **Then**, all notifications when away from site have been retrieved.
- **Given**, I am on my user settings page **When**, I update one of the travel notification settings **Then**, it will be reflected immediately and persistently.
- **Given**, I am part of a trip and logged in **When**, a trip notification tripping event happens **Then**, I will receive a notification on the site.

User Story #11: Remove Activity Options

"As a user, I would like to be able to click a button to remove options."

- **Owner:** Shashank Mukkera
- **Estimated Hours:** 7 Hours
- **Tasks:**
 - Extend the itinerary data model for generator-backed options: either add `status: "candidate" | "removed" | "selected"` (and optional `optionGroupId`) on embedded itinerary items in `Trip`, or store dismissed option IDs on `TravelGroup` / `CalendarEvent` metadata so removals persist across reloads. (1.5 hrs)
 - Build DELETE or PATCH `/api/groups/{groupId}/itinerary/options/{optionId}` (or `/api/trip/.../options/{optionId}`) to mark an option as removed: validate session, ensure the user is a group member with edit rights, and reject if the option is already finalized on the calendar. (1.5 hrs)
 - Add a "Remove" (or dismiss) control on each candidate activity card in the itinerary generator UI; on success, remove the card from the list with optimistic UI and rollback on error. (1.5 hrs)

- Ensure removed options are excluded from the next "generate" or "regenerate" pass by passing `excludedOptionIds` (or names) into the generator pipeline, consistent with existing `avoidActivities` / `avoidLocations` on `Trip`. (1 hr)
- Write unit tests: member with edit permission can remove; viewer cannot (403); removed option does not reappear after refresh; invalid optionId returns 404. (1.5 hrs)
- **Acceptance Criteria:**
 - Given that I am a group member allowed to edit the itinerary, When I click "Remove" on a generated candidate option, Then that option is dismissed and no longer appears in the candidate list.
 - Given that I have removed an option, When I reload the page, Then the option stays removed (persisted server-side).
 - Given that I have only view access, When I try to remove an option, Then the action is rejected (403).
 - Given that an activity is already committed to the shared calendar as a final event, When I try to remove it via the candidate flow, Then the API returns a clear error (e.g. 400) explaining the item is not a dismissible option.

User Story #12: Vote on Activity Options

"As a user, I would like to be able to vote on specific activity options created by the itinerary generator."

- **Owner:** Shashank Mukkera
- **Estimated Hours:** 9 Hours
- **Tasks:**
 - Add a Mongoose sub-schema or collection for option groups and votes (e.g. `optionGroupId`, `groupId`, `candidates[ ]`)

with `optionId`, `title`, `placeId`; votes as { `userId`, `optionId` }[] or a Map; optional `closesAt`). (2 hrs)

- Build POST `/api/groups/{groupId}/itinerary/votes` to cast or change a vote (one vote per user per option group); validate membership, Zod body { `optionGroupId`, `optionId` }. (1.5 hrs)
- Build GET `/api/groups/{groupId}/itinerary/votes` (or embed in group itinerary GET) to return tallies and the current user's selection per group. (1 hr)
- Build optional POST `/api/groups/{groupId}/itinerary/votes/{optionGroupId}/finalize` for leaders to promote the winning option to a `CalendarEvent` (or link to existing apply flow). (1.5 hrs)
- Create the voting UI: radio buttons or upvote-style controls on each candidate in a group, live vote counts, and a "Group choice" badge on the leading option; use SWR to refresh tallies after voting. (2 hrs)
- Write unit tests: one vote per user per group; vote changes update tally; non-members 403; finalize only for leader (if implemented). (1 hr)

- **Acceptance Criteria:**

- Given that I am a group member, When I vote for one activity option in a generated set, Then my vote is recorded and the tally updates for everyone.
- Given that I change my vote to a different option in the same group, Then my previous vote is replaced and tallies reflect the change (still one vote per user per group).
- Given that I am not in the group, When I attempt to vote, Then the request is rejected (403).
- Given that the leader finalizes the poll (if this sub-feature is included), When they confirm, Then the winning option becomes a scheduled itinerary/calendar item according to your product rules.

User Story #13: Export Itinerary to Calendar

"As a user, I would like to be able to export itineraries to my preferred calendar."

- **Owner:** Shashank Mukkera
- **Estimated Hours:** 8.5 Hours
- **Tasks:**
 - Build GET `/api/groups/{groupId}/itinerary/export.ics` that serializes `CalendarEvent` documents for the group into a valid iCalendar (RFC 5545) file (`VCALENDAR / VEVENT`), with `UID`, `DTSTART/DTEND`, `SUMMARY`, `LOCATION`, `DESCRIPTION`, and proper escaping. Support optional query params: `from`, `to`, `includeManual`, `includeItinerary`. (2.5 hrs)
 - Add a "Subscribe" variant: same feed URL with `?token=` for a long-lived opaque token stored on the user or group (security review) or document that export is session-only and calendar apps re-download on sync — pick one and implement minimally. (1.5 hrs)
 - Build GET `/api/groups/{groupId}/itinerary/export/google` that returns a `https://calendar.google.com/calendar/render?action=TEMPLATE&...` URL for adding the next event or bulk events (or document one-event-at-a-time vs. full `.ics`). (1 hr)
 - Create the Export UI: dropdown "Add to Google Calendar" / "Download .ics" / "Copy subscription link" with Zod-validated date range; show success toast and filename `boilerbridge-{groupName}.ics`. (2 hrs)
 - Write unit tests: `.ics` parses with correct event count; empty itinerary returns empty calendar or informative message;

non-member 403; timezone handling documented (UTC vs. local). (1.5 hrs)

- **Acceptance Criteria:**

- Given that I am a group member, When I click "Download .ics," Then I receive a file that imports successfully into Apple Calendar, Google Calendar, or Outlook.
- Given that I choose "Add to Google Calendar," When I confirm, Then I am taken to Google with fields pre-filled for at least one event (or the documented behavior for bulk export).
- Given that I am not a member, When I hit the export endpoint, Then I receive 403 Forbidden.
- Given that I filter by date range in the export UI, When I download, Then only events in that range appear in the file.

User Story #14: Filter by Accessibility Requirements

"As a user, I would like to be able to filter generated locations based on any accessibility requirements."

- **Owner:** Shashank Mukkera
- **Estimated Hours:** 8 Hours
- **Tasks:**
 - Define an accessibility profile model: store on `User` or `Trip / TravelGroup` — e.g. `wheelchairAccessible`, `stepFree`, `accessibleRestroom`, `hearingAssistance`, `visualAssistance` (booleans or enum set). Validate with Zod. (1.5 hrs)
 - Extend the Places search / itinerary generator inputs (`app/api/activities/search`, group itinerary generate) to pass Google Places accessibility-related fields where available (e.g. wheelchair-accessible entrance) and filter out candidates that do not match the profile. (2.5 hrs)

- Add UI controls: multi-select or checklist "Accessibility needs" on trip setup and/or group itinerary sidebar; persist and reload with the trip/group. (1.5 hrs)
- When displaying search or generated results, show accessibility badges on cards and a client-side filter toggle to hide non-matching places (defense in depth if API data is incomplete). (1.5 hrs)
- Write unit tests: filter removes mismatched mock places; empty results shows a helpful message; API rejects invalid profile payloads (400). (1 hr)
- **Acceptance Criteria:**
 - Given that I set accessibility requirements before generating or searching, When results load, Then I only see locations that satisfy those requirements (or are explicitly unknown with a warning, per your chosen strictness — document in UI).
 - Given that I change my accessibility filters, When I apply them, Then the result list updates without a full page reload (or with a clear refresh pattern).
 - Given that no places match my filters, When I search or generate, Then I see an empty state explaining that no matching venues were found.
 - Given that I submit invalid accessibility payload, When the API validates, Then I receive 400 with field-level errors.

User Story #15:

"As a user, I would like to be able to post itineraries visible to all app users."

- **Owner:** Shreyansh Tehanguria
- **Estimated Hours:** 11
- **Tasks:**

- Create a **PublicItinerary** schema or extend existing **Trip** schema with **isPublic**, **visibility**, and metadata fields (likes, views, ownerId). (2 hrs)
- Build **POST /api/itineraries/publish** to allow users to publish their itinerary with validation (authenticated user, valid itinerary). (1.5 hrs)
- Build **GET /api/itineraries/public** to fetch all public itineraries with pagination and sorting (latest, trending). (2 hrs)
- Implement access control to ensure only owners can toggle visibility and edit their public itineraries. (1 hr)
- Create Public Itinerary Feed UI showing cards with itinerary preview (location, duration, highlights). (2 hrs)
- Add interaction features like “view details” and basic engagement tracking (views increment). (1.5 hrs)
- Write unit and integration tests for publishing, fetching, and authorization flows. (1 hr)
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- **Acceptance Criteria:**
 - Given I have created an itinerary, When I click “Publish,” Then it becomes visible in the public feed.
 - Given I am viewing the public feed, When I open an itinerary, Then I can see its details but cannot edit it.
 - Given I am not the owner, When I try to modify a public itinerary, Then the system returns a 403 error.
 - Given I unpublish an itinerary, When I refresh the public feed, Then it no longer appears.

User Story #16:

""As a user, I would like to be able to make changes to only one part of an itinerary."

- **Owner:** Shreyansh Tehanguria
- **Estimated Hours:** 12
- **Tasks:**

- Extend itinerary schema to support section-level updates (e.g., day-wise or activity-level IDs). (2 hrs)
 - Build `PATCH /api/itinerary/:id/section` endpoint to update only a specific part (day/activity). (2 hrs)
 - Add validation logic to ensure only modified sections are updated while preserving the rest. (1.5 hrs)
 - Implement optimistic UI updates for partial edits without full reload. (1.5 hrs)
 - Update frontend UI to allow editing individual sections (edit button per activity/day). (2 hrs)
 - Ensure concurrency handling (versioning or last-write-wins strategy for edits). (1.5 hrs)
 - Write unit and integration tests for partial updates and conflict handling. (1.5 hrs)
- **Acceptance Criteria:**
 - Given I want to modify a single activity, When I edit it, Then only that activity is updated without affecting others.
 - Given I refresh the page, When I view the itinerary, Then the updated section persists correctly.
 - Given multiple users edit different sections, When updates occur, Then changes do not overwrite unrelated sections.
 - Given I am not authorized, When I try to edit a section, Then the system rejects the request with a 403 error.

User Story #17:

"As a user, I would like to be able to have an offline itinerary view. "

- **Owner:** Shreyansh Tehanguria
- **Estimated Hours:** 11
- **Tasks:**
 - Implement caching strategy using browser storage (IndexedDB or localStorage) for itinerary data. (2 hrs)

- Build service worker setup to enable offline access to saved itineraries. (2 hrs)
- Create logic to sync cached itinerary data when user comes back online. (1.5 hrs)
- Add UI indicator for offline mode (banner or badge). (1 hr)
- Modify itinerary fetch logic to fallback to cached data when API is unavailable. (1.5 hrs)
- Ensure data consistency and invalidation strategy when itinerary updates occur. (1.5 hrs)
- Write unit and integration tests for offline access and sync behavior. (2 hrs)
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- **Acceptance Criteria:**
 - Given I have previously viewed an itinerary, When I lose internet connection, Then I can still access it offline.
 - Given I am offline, When I open the itinerary page, Then cached data is displayed instead of an error.
 - Given I come back online, When the app reconnects, Then the itinerary syncs with the latest server data.
 - Given no cached data exists, When I try to access offline, Then the UI shows an appropriate error state.

User Story #_:

"As a user, "

- **Owner:**
- **Estimated Hours:**
- **Tasks:**
 -
- **Acceptance Criteria:**
 - **Given, When, Then**

3. Backlog (Future Sprints)

User Account

- ~~● As a user, I would like to be able to create an account.~~
- ~~● As a user, I would like to be able to login and manage my account settings.~~
- ~~● As a user, I would like to be able to edit my profile details, like name, school, location, and profile picture.~~
- ~~● As a user, I would like to be able to reset my password via email/sms if I forget it.~~
- ~~● As a user, I would like to be able to link my external calendar (Google/Outlook) to sync trip dates.~~
- ~~● As a user, I would like to be able to verify my student status (.edu email) to unlock student filters.~~
- ~~● As a user, I would like to be able to manage my notification preferences (Email, SMS, Push) for trip updates.~~

Travel Groups

- ~~● As a user, I would like to be able to create travel groups.~~
- ~~● As a user, I would like to be able to manage my travel group.~~
- ~~● As a user, I would like to be able to leave my travel group.~~
- ~~● As a user, I would like to be able to access a shared calendar for my travel group.~~
- ~~● As a group leader, I would like to be able to transfer my leader's permission to another member.~~
- As a group leader, I would like to be able to remove specific people from the travel group.
- ~~● As a user, I would like to be able to message the members of my travel group.~~

Trip Creation

- ~~● As a user, I would like to be able to input my travel preferences.~~
- ~~● As a user, I would like to create a budget for the trip.~~
- ~~● As a user, I would like to be able to specify transportation preferences.~~
- ~~● As a user, I would like to be able to add activities/locations as must haves.~~
- ~~● As a user, I would like to be able to add activities/locations to avoid.~~
- ~~● As a user, I would like to be able to generate a random location based on budget.~~
- ~~● As a user, I would like to be able to see recommendations/extra info for activities.~~

Itinerary Interaction/Modification

- ~~● As a user, I would like to be able to see reviews for places people have travelled.~~
- ~~● As a user, I would like to be able to see a summary of reviews for activities.~~
- ~~● As a user, I would like to be able to visit sites for more information about activities.~~
- ~~● As a user, I would like to be able to go to activity booking sites via external links.~~
- ~~● As a user, I would like to be able to find temporary bag storage.~~
- ~~● As a user, I would like to be able to have an offline itinerary view.~~
- As a user, I would like to be able to see a visual map representation of the itinerary.
- As a user, I would like to be able to have an offline map view.
- ~~● As a user, I would like to be able to create reminders for my travels.~~
- ~~● As a user, I would like to be able to access an SOS button that shows emergency numbers.~~
- ~~● As a user, I would like to be able to write reviews for places I've visited.~~
- ~~● As a user I would like to be able to create "Polls" within the group to vote on destination options.~~

- ~~As a user, I would like to be able to assign roles (Admin, Viewer) to group members.~~
- ~~As a user, I would like to be able to view a "Group Board" for pinned announcements.~~
- As a user, I would like to be able to generate alternative plans.
- ~~As a user, I would like to be able to make changes to only one part of an itinerary.~~
- ~~As a user, I would like to be able to click a button to regenerate specific parts of the itinerary.~~
- ~~As a user, I would like to be able to click a button to remove options.~~
- ~~As a user, I would like to be able to vote on specific activity options created by the itinerary generator.~~
- As a user, I would like to be able to retrieve previous itineraries.
- ~~As a user, I would like to be able to post itineraries visible to all app users.~~
- ~~As a user, I would like to be able to share itineraries via link.~~
- ~~As a user, I would like to be able to export itineraries to my preferred calendar.~~
- ~~As a user, I would like to be able to select specific "Vibe" tags (Nightlife, Nature, Historical) to guide the itinerary generator.~~
- As a user, I would like to be able to lock specific activities I like so they don't change during regeneration.
- As a user, I would like to be able to drag and drop activities to reorder the schedule visually.
- ~~As a user, I would like to generate a "Rainy Day" alternative for my plans.~~
- ~~As a user, I would like to be able to filter generated locations based on any accessibility requirements.~~
- ~~As a user, I would like to be able to view a 10-day weather forecast in the itinerary.~~

Itinerary Budget

- ~~As a user, I would like to be able to post a shared cost to the group ledger.~~

- ~~● As a user, I would like to be able to split the costs with my group.~~
- ~~● As a user, I would like to be able to view a list of what everyone owes at the end of the trip.~~
- ~~● As a user, I would like to be able to request payment from specific group members for a shared cost.~~
- ~~● As a user, I would like to be able to send a payment confirmation notice to a member after paying them back.~~

Communication

- ~~● As a user, I would like to be able to connect with new people by finding and adding them as friends.~~
- ~~● As a user, I would like to be able to create a shared photo album for trips.~~
- ~~● As a user, I would like to be able to send text notifications to my group.~~
- ~~● As a user, I would like to be able to send email notifications to my group.~~
- As a user, I would like to be able to send in-app notifications to my group.
- As a user, I would like to be able to receive text notifications about my travel.
- As a user, I would like to be able to receive email notifications about my travel.
- ~~● As a user, I would like to be able to receive in-app notifications about my travel.~~
- As a user, I would like to be able to share my real-time location with added friends.